

UNIVERSIDADE D
COIMBRA

Tiago Silva

Towards Fair Algorithmic Recourse with Kolmogorov-Arnold Networks

Dissertation in the context of the Master in Informatics Engineering, specialization in Artificial Intelligence, advised by Professor Pedro Henriques Abreu and Professor Bruno Veloso and presented to the Department of Informatics Engineering of the Faculty of Sciences and Technology of the University of Coimbra.

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Rumo a um Recurso Algorítmico Justo com Redes de Kolmogorov-Arnold

Dissertação no âmbito do Mestrado em Engenharia Informática, especialização em Inteligência Artificial, orientada pelo Professor Pedro Henriques Abreu e pelo Professor Bruno Veloso e apresentada ao Departamento de Engenharia Informática da Faculdade de Ciências e Tecnologia da Universidade de Coimbra.

Janeiro 2027

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I would like to thank...

Statement on the use of artificial intelligence tools

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Abstract

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Keywords

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Chapter 1

Introduction

1.1 Context and Motivation

1.2 Objectives and Research Questions

1.3 Work Plan and Risk Analysis

1.3.1 Work Plan

1.3.2 Risk Analysis and Mitigation

1.4 Document Structure

Chapter 2

State of the Art

2.1 Background Knowledge

2.1.1 Notation

This subsection fixes the notation used in the rest of the document. Uppercase italic letters denote random variables, such as a feature X_j, the outcome Y or the protected attribute A, and the corresponding lowercase letters denote the values they take (x_j, y, a). Bold letters denote vectors or sets of variables: $\mathbf{X} = (X_1, \ldots, X_d)$ collects the d features and $\mathbf{x}$ holds the feature values of one person. Calligraphic letters denote graphs and sets of indices, such as the causal graph $\mathcal{G}$; $\mathcal{N}$ and Unif denote the normal and uniform distributions. The index j refers to the variable being explained, i, k and l to its parents or to intervened variables, m to a basis function and t to a time window. A hat marks a quantity estimated from data, as in $\hat{f}_j$ for the estimate of f_j. The letter σ has two uses, kept apart by their form: $\sigma(z)$, always written with an argument, is the logistic function, and σ_j^2, always written with a subscript, is the noise variance of X_j.

Table 2.1 lists the symbols that are used in more than one section, together with the section that defines each of them. Symbols used in a single place are defined where they appear.

2.1.2 Counterfactual Explanations and Algorithmic Recourse

2.1.3 Structural Causal Models and Interventions

Recourse asks what would happen to a person's decision if they acted, and counterfactual fairness asks what the decision would have been had a protected attribute been different. Both are, inevitably, causal questions, and this thesis answers them with structural causal models (SCMs) [Pearl, 2009a; Peters et al., 2017]. This subsection introduces SCMs and interventions.

Symbol	Meaning	Section
People, decisions and recourse		
$\mathbf{x} = (x_1, \ldots, x_d)$	feature values of one person	2.1.2
X_j, Y	j-th feature; outcome	2.1.2
h	classifier (decision model)	2.1.2
$\mathbf{x}'$	counterfactual explanation of $\mathbf{x}$	2.1.2
$c(\mathbf{x}, \cdot)$	cost of an action for the person $\mathbf{x}$	2.1.2
A, a	protected attribute and its value	2.1.4
Causal models		
$\mathcal{G}$	causal graph	2.1.3
$\mathbf{PA}_j$, $\mathbf{pa}_j$	parents of X_j and their values	2.1.3
f_j, $\hat{f}_j$	structural equation of X_j and its estimate	2.1.3
U_j, u_j	noise of X_j and its value	2.1.3
$\mathrm{do}(X_i = \theta)$	intervention that sets X_i to θ; $\mathrm{do}(\mathbf{X}_\mathcal{I} = \boldsymbol{\theta})$ sets every X_i with $i \in \mathcal{I}$	2.1.3
$\sigma(z)$	logistic function, $1/(1+e^{-z})$	2.1.3
Change over time		
t, P_t	time window; distribution of the data in window t	2.1.5
Curves and the causal KAN		
B_m, β_m	m-th B-spline basis function and its height (coefficient)	2.1.6
α_j	constant term in the equation of X_j	2.1.6
ϕ_{ij}	curve on the edge $X_i \to X_j$	2.1.6
ψ_{klj}	surface for the pair of parents X_k and X_l of X_j	2.1.6
λ	strength of the smoothness belief	2.1.7
σ_j^2	noise variance of X_j	2.1.7

Table 2.1: Symbols used in more than one section, with the section that defines each.

Structural causal models. An SCM describes how each variable is produced from its direct causes. Following Peters et al. [2017], an SCM over $X_1, \ldots, X_d$ consists of d structural assignments

$$X_j := f_j(\mathbf{PA}_j, U_j), \qquad j = 1, \ldots, d, \tag{2.1}$$

where $\mathbf{PA}_j \subseteq \{X_1, \ldots, X_d\} \setminus \{X_j\}$ are called the parents of X_j, together with a joint distribution $P_{\mathbf{U}}$ over the noise variables $U_1, \ldots, U_d$, which are required to be jointly independent. Peters et al. [2017] write the noise variables as N_j. This thesis writes U_j, the letter used by Pearl [2009b] and the recourse literature [Karimi et al., 2021], where U_j is called an exogenous variable and X_j an endogenous one. There, independence is an extra assumption rather than part of the definition: Pearl [2009b] calls a model whose graph is acyclic and whose exogenous variables are jointly independent *Markovian*. Drawing an arrow from each parent to its child gives the causal graph $\mathcal{G}$, which is assumed to be a directed acyclic graph. The parents of X_j are its direct causes and U_j gathers everything else that affects X_j but is not modelled; requiring the noise variables to be independent and, therefore, rules out hidden variables that affect two observed ones (causal sufficiency). An acyclic SCM entails a single joint distribution over $X_1, \ldots, X_d$ [Peters et al., 2017], which factorises into one conditional distribution per variable [Pearl, 2009a]:

$$P(X_1, \ldots, X_d) = \prod_{j=1}^{d} P(X_j \mid \mathbf{PA}_j). \tag{2.2}$$

Each factor is the *causal mechanism* of one variable. The principle of independent mechanisms states that these factors do not inform or influence one another, so that one of them can change while the others stay the same [Peters et al., 2017]. Section 2.1.5 uses this idea to describe drift as a change in some mechanisms but not in others.

Running example. Figure 2.1 shows the SCM used as a running example. A person's income X_1 affects their savings X_2, and both affect the bank's decision Y, where $Y = 1$ means that the loan is approved:

$$\begin{aligned} X_1 &:= U_1, \\ X_2 &:= 0.5\, X_1^2 + U_2, \\ Y &:= \mathbf{1}[\, U_Y < \sigma(-4 + X_1 + 0.5\, X_2)\,], \end{aligned} \tag{2.3}$$

with $U_1 \sim \text{Unif}(0, 4)$, $U_2 \sim \mathcal{N}(0, 0.3^2)$ and $U_Y \sim \text{Unif}(0, 1)$, where $\sigma(z) = 1/(1 + e^{-z})$ is the logistic function and $\mathbf{1}[\cdot]$ equals 1 when its condition holds and 0 otherwise. The last equation means that a person is approved with probability $\sigma(-4 + X_1 + 0.5\, X_2)$. The noise U_2 can be read as a saving habit: a positive value means that the person saves more than others with the same income.

Interventions. An intervention sets a variable from outside the system. The intervention $\text{do}(X_i = \theta)$ replaces the equation of X_i with the constant θ and leaves every other equation unchanged [Pearl, 2009a]:

$$X_i := \theta, \qquad X_j := f_j(\mathbf{PA}_j, U_j) \quad \text{for all } j \neq i. \tag{2.4}$$

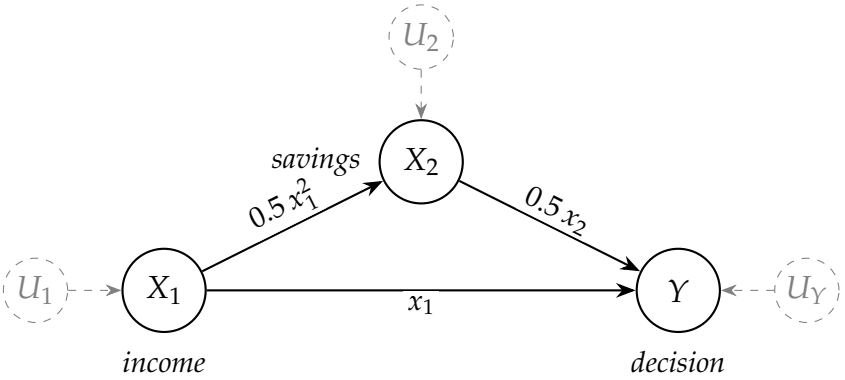


Figure 2.1: Causal graph of the running example (Equation 2.3). Solid arrows are causal effects, labelled with the term that each parent adds to its child's equation; dashed arrows come from the noise variables.

The result is a new SCM, whose graph has no arrows into X_i and whose distribution is the *interventional* distribution $P(X_1, \ldots, X_d \mid \mathrm{do}(X_i = \theta))$. Several variables can be set at once: $\mathrm{do}(\mathbf{X}_{\mathcal{I}} = \boldsymbol{\theta})$ replaces the equation of every X_i with i in the set of indices $\mathcal{I}$. Intervening is not the same as conditioning. In the running example, $P(Y \mid X_2 = x_2)$ describes the people who happen to have savings x_2, and those with high savings tend to also have a high income, whereas $P(Y \mid \mathrm{do}(X_2 = x_2))$ describes what happens when savings are set to x_2 and income stays as it was. In algorithmic recourse, the action a person takes is modelled as an intervention on the features they can change [Karimi et al., 2021]; Section 2.2.2.2 reviews this view.

2.1.4 Fairness: Statistical and Causal Notions

2.1.5 Concept Drift and Distribution Shift

2.1.6 Splines and Additive Models

2.1.7 Bayesian Smoothing and Model Evidence

2.1.8 Kolmogorov–Arnold Networks

2.2 Literature Review

This chapter reviews the work on which a causal, fair and drift-aware recourse method can build. Section 2.2.1 covers how counterfactuals are searched for, and Section 2.2.2 how they are kept plausible and causally consistent. Section 2.2.3 asks whether recourse is fair, and Section 2.2.4 examines what happens when models, populations or causal mechanisms change over time. Finally, Section 2.2.8 turns to readable representations of structural equations, including those based

on Kolmogorov–Arnold Networks (KANs).

2.2.1 Counterfactual Search Methods

In this section, we analyse the current landscape of counterfactual search methods, focusing on which strategies are used to find valid counterfactuals and which trade-offs they make between search efficiency, plausibility and causal consistency. We group the methods into three categories: exact solver-based methods, gradient-based methods, and evolutionary and surrogate-based methods.

2.2.1.1 Exact Solver-Based Methods

Russell [2019] proposes a mixed-integer programming approach to generating counterfactual explanations for linear classifiers, including logistic regression, linear regression and support vector machines. The search is formulated as an optimisation problem that minimises a weighted ℓ_1 distance between the original instance and the counterfactual, which keeps changes sparse (i.e., as few features as possible are changed). Its main contribution is a set of constraints, called a mixed polytope, that guarantees coherent counterfactuals: solutions that map back onto the underlying data structure, even when a variable can take either a continuous value or one of several discrete states. Diversity is obtained by restricting the variables altered in previously generated counterfactuals. Results on the LSAT and FICO datasets show that the method finds counterfactuals that are close to the original instance and cross the model's decision boundary, while avoiding the brute-force enumeration required by earlier approaches. However, the approach is designed for linear models, so it does not extend directly to complex neural networks. It is also blind to causal relationships between features: it guarantees coherence, not causal consistency. Finally, explanations are presented as short text statements, and the author notes that a complete set can be overwhelming and that more work is needed on how to present and navigate it interactively.

Kanamori et al. [2020] also rely on mathematical optimisation, using mixed-integer linear optimisation for linear models and tree ensembles, but do so in a distribution-aware manner. Their cost function combines the Mahalanobis distance, which accounts for feature correlations, with the local outlier factor, which avoids counterfactuals in low-density regions. However, this comes at a computational price: on the FICO dataset with a random forest, DACE took 519 seconds on average under a 600-second limit, against 22 to 34 seconds for the baselines. Moreover, although it encourages counterfactuals not to break correlations, it does not guarantee causal consistency.

Karimi et al. [2020a] propose MACE, a model-agnostic method that generates counterfactuals using Satisfiability Modulo Theories (SMT) solvers. The model, the distance function and the constraints are all translated into logic formulae, and additional constraints make the counterfactuals plausible and diverse. The method finds a counterfactual for every individual considered (100% coverage)

at provably optimal distances. The problem is that the model must be expressible in first-order logic, which may not be feasible for all types of models, especially complex ones. Once again, the method is blind to causal relationships between features, and it can be computationally expensive because its running time depends directly on the solver: under five seconds for logistic regression, but between one minute and three hours for a multilayer perceptron.

2.2.1.2 Gradient-Based Methods

Van Looveren and Klaise [2020] use class prototypes to guide the counterfactual search. A term added to the loss function encourages the counterfactual to be close to the prototype of the desired class, obtained either from an encoder or from class-specific k-d trees, which makes the generated counterfactuals more interpretable and keeps them closer to the data manifold. Evaluation on the MNIST and Breast Cancer Wisconsin datasets shows that the method generates counterfactuals that are close to the original instance and cross the decision boundary, and that it is faster than searches without prototypes. It also removes the bottleneck of numerical gradient evaluation for black-box models. Notwithstanding, the method relies heavily on the quality of the prototypes and is, again, blind to causal relationships between features. Matching a prototype can also lead to less sparse counterfactuals, as many features may have to change, and the authors report a trade-off between sparsity and interpretability.

Mothilal et al. [2020] propose DiCE, a framework that generates multiple, diverse and actionable counterfactuals for machine learning classifiers. It moves explainability from showing a single path of recourse to offering individuals a variety of options to choose from. DiCE initialises k candidates randomly and optimises them using a determinantal point process (DPP) term to ensure diversity, along with proximity to the original instance. The novelty of the search mechanism is that it optimises all candidates jointly. The framework is evaluated on the COMPAS, Adult-Income, German-Credit and Lending Club datasets. Its limitations are that it assumes a differentiable and static model, and that it varies features independently, which can produce causally infeasible counterfactuals; causal constraints are only used to filter these out afterwards.

2.2.1.3 Evolutionary and Surrogate-Based Methods

White and d'Avila Garcez [2019] introduce CLEAR and a definition of the fidelity of local explanation models to the underlying classifier, based on distances to a decision boundary. The method generates b-counterfactuals, which state the minimum changes necessary to flip the classifier's prediction, and uses them to measure and improve the fidelity of local regression models. Across five case studies, CLEAR improves on the fidelity of LIME by an average of over 40%. However, its neighbourhood datasets do not always represent the relevant regions of the feature space, fidelity varies considerably between datasets, and b-counterfactuals that change several features are left as future work.

Dandl et al. [2020] propose MOC which, to the best of the authors' knowledge, is the first method to formalise the counterfactual search as a multi-objective optimisation problem. It adapts the NSGA-II algorithm to return a set of Pareto-optimal counterfactuals with different trade-offs between four objectives: reaching the desired prediction, staying close to the original instance, changing few features and staying close to the observed data. The method is model-agnostic and can be applied to any black-box model. Nevertheless, the benchmark is limited to binary classification on datasets with at most 40 features, which raises concerns about scalability to high-dimensional data. The work also leaves open how one should choose the final counterfactual from the Pareto front, gives no guarantee of causal consistency, and can be computationally expensive due to the nature of evolutionary algorithms.

2.2.2 Plausibility and Causal Consistency

A counterfactual that is close to the original instance is not necessarily one that a person can reach. This section reviews two responses to that problem: keeping counterfactuals on the data manifold, and explicitly modelling the causal relationships between features.

2.2.2.1 Data-Manifold Methods

Poyiadzi et al. [2020] propose FACE, a graph-based method for generating counterfactuals. It constructs a graph where nodes are observed data points (e.g., people's financial profiles), edges connect nearby nodes, and density-weighted edge weights represent the cost of moving from one node to another, which favours paths through high-density regions. The method then searches for the shortest path from the original instance to a node of the desired class that satisfies prediction and density thresholds. On a synthetic dataset and on MNIST, the authors show that the method generates not only counterfactuals but also intermediate steps that explain how to reach them. However, the method does not guarantee that real-world actions will succeed: it ensures plausibility, which should not be confused with causality, nor with proof that an individual can take that set of actions. Evaluation on real-world data of a dynamic nature is left as future work.

Olson et al. [2021] introduce counterfactual state explanations for deep reinforcement learning agents via generative deep learning. An encoder maps the game state to a latent representation, which is fed together with the agent's action distribution into a generator. The result is the minimal visual modification that makes the agent choose a different action. The networks are trained using a combination of adversarial and reconstruction losses, so that the counterfactual states are both realistic and close to the original state. Two user studies on Atari games with non-experts show that (i) the generated states have sufficient fidelity to the actual game, and (ii) they help participants identify a flawed agent, which the authors deliberately blinded to its own ship's position, more effectively than a nearest-neighbour baseline or no explanation at all. The method could therefore serve as an auditing tool to identify agent misbehaviour before real-world de-

ployment. Beyond being restricted to reinforcement learning agents and evaluated on Atari games rather than on high-stakes real-world problems, the method does not explain the long-term effects of the chosen action on the environment, so no temporal consistency is guaranteed. The generative model also has no concern for causal consistency, which increases the risk of providing misleading explanations.

Joshi et al. [2019] introduce REVISE, a method for plausible algorithmic recourse through generative models. A variational autoencoder (VAE) encodes a person into a latent space, where gradient descent pushes the candidate towards the desired label while keeping it close to the original instance. Recourse is the difference between the original and the generated instance. Identified limitations include that the suggested changes are not sparse, which may make them difficult for a person to act upon; that the reported evaluation is mainly qualitative, with quantitative results relegated to the appendix; and that realism depends on the VAE. Although the authors use a method (CEVAE) that captures the effect of a treatment on the outcome adjusted for confounders (variables that influence both the treatment and the outcome), they do not model causal relationships between features, so recourse is still a list of independent feature changes, and they provide no guarantee that it works. They also show that recourse can act as a diagnostic, revealing whether a classifier is systematically confounded by specific attributes.

2.2.2.2 Causal Constraints and Recourse as Interventions

Mahajan et al. [2019] are among the first to connect the feasibility of counterfactuals with causal relationships between features, arguing that feasibility is fundamentally a causal concept. They use a VAE-based generator with constraints derived from a partial structural causal model (SCM) and, when such constraints cannot be expressed, learn feasibility from users' binary feedback. However, causal relationships are soft constraints and are therefore not guaranteed to be satisfied, the output is still a counterfactual state rather than a set of actions, and the comparison covers only one external baseline.

Karimi et al. [2021], in contrast, present a method that outputs actions as interventions on an SCM, thus guaranteeing causal consistency when the SCM is correct. They prove that acting on a nearest counterfactual explanation is only guaranteed to achieve recourse when the acted-upon variables have no descendants, and that a minimal intervention never costs more than a counterfactual-based action that achieves recourse. The method extends the aforementioned MACE framework and distinguishes immutable, mutable but non-actionable, and actionable features, successfully reframing recourse from a nearest-point search to actions and their consequences. Regarding limitations, the method requires the true SCM, with no hidden confounders and invertible equations, and the experiments are limited to small graphs, with little robustness analysis. The authors leave open (i) actions that affect multiple features, (ii) sequential actions and (iii) the choice of cost functions.

Karimi et al. [2020b] tackle the weakness of having to know the true SCM. They start by showing that two causal models that are observationally equivalent can lead to different recourse recommendations, so unless we know the true structural equations, we cannot guarantee that recourse is valid, no matter how much data is available. They then propose two probabilistic methods for the case where the structural equations are unknown but the causal graph is known: individual recourse using a Gaussian process SCM (GP-SCM), and subpopulation recourse based on the conditional average treatment effect (CATE). The first assumes that each structural equation is a function of its parents plus Gaussian noise, yielding a distribution over a person's counterfactuals. The second drops all assumptions about the structural equations and estimates the average effect of the action on people who share the same non-descendant features. The limitations are that (i) the causal graph is still required, as only the equations are relaxed, (ii) the additive Gaussian noise assumption may not hold under the true SCM, leading to sub-optimal solutions, (iii) the experiments are small, and (iv) subpopulation recourse only ensures that the average effect of the action is positive for a group of people, not that it works for a given individual. The authors treat the classifier as given, and they point to fair decision making with a classifier that evolves over time as an important direction for future research.

2.2.3 Fairness of Recourse

Even when recourse is valid, it may be harder for some people to achieve than for others. This section reviews how that inequality has been measured, first through distance-based audits, then through causal notions of fairness, and finally over time.

Sharma et al. [2020] propose CERTIFAI, a framework that uses counterfactual explanations generated by a genetic algorithm as a model-agnostic way of auditing machine learning models for fairness, robustness and interpretability. They introduce CERScore, a black-box robustness metric defined as the average distance between inputs and their counterfactuals, and burden, the expected distance to the counterfactual for a group, which allows groups to be compared. The main limitations are that a genetic approach does not guarantee optimality, that actionability depends on feature constraints set by the user, and that causal consistency is not considered.

von Kügelgen et al. [2022] make recourse a fairness criterion and connect it formally to counterfactual fairness. They define group-level and individual-level notions of fair causal recourse and show that: (i) group-level fairness does not imply individual-level fairness; (ii) a counterfactually fair classifier does not guarantee fair recourse (fair prediction and fair recourse are complementary, not substitutes); and (iii) if the classifier only uses features that are non-descendants of the protected attribute, and the feasible actions and their costs do not change under a counterfactual change of that attribute, then recourse is individually fair. They also show that non-causal metrics misrepresent recourse unfairness, and they detect recourse discrimination in a case study on the Adult dataset. Regarding limitations, the individual-level notion needs a counterfactual on the pro-

tected attribute, which requires the full SCM; the fix based on non-descendant features can lead to a drop in accuracy; constraining training directly is difficult because the search over intervention targets is combinatorial; and the individual-level measure is a maximum over individuals, so it may be dominated by outliers. As changing the classifier does not address the root of the problem, the authors also propose societal interventions on the data-generating process as an alternative.

Fonseca et al. [2023] move the question to a dynamic setting. In agent-based simulations, they define a recourse reliability metric and study two effects: competition with other agents acting on recourse, and competition with new agents entering the environment. Only a small set of parameter settings yields recourse that is reliable over time. However, these results rest on simulated adaptation functions, which only approximate human behaviour, and no causal model is used.

Bell et al. [2025] build on this simulation work and define two new metrics, grounded in substantive equality of opportunity, for measuring fairness in this context: Effort-to-Recourse (ETR) and Time-to-Recourse (TTR). Disadvantaged groups need more of both, even under a fair classifier. The proposed Circumstance-Normalized Selection (CNS) adjusts selection after scoring to reduce that gap. In terms of limitations, the work relies on agent simulations, so its results depend on assumed human behaviour, it uses no causal models, and the experiments use a simple logistic classifier. The authors also point to the lack of public real-world recourse data as a main limitation.

2.2.4 Recourse and Causal Models Under Drift

Most of the works above assume that the model, the population and the causal mechanisms stay fixed. This section relaxes that assumption in three steps: the world changes and invalidates recourse, recourse itself causes change, and causal structure has to be learned from change.

2.2.4.1 Recourse in a Changing World

One of the most relevant works on recourse in changing environments is De Toni et al. [2025], where the authors treat the world as a causal process that changes over time, rather than treating the model as the only shifting component. They start by proving that counterfactual recourse for a future time cannot be optimal unless the noise has zero variance, and that robust uncertainty sets are not enough (i.e., searching for a counterfactual and giving it an error margin does not counteract time). Their solution, T-SAR, gives subpopulation recourse using an estimator of the stochastic process. Notwithstanding, the estimator is fitted once and the classifier is also trained once, so trends in the classifier and in the cost are not studied, and the results depend on the quality of the estimator. The work covers one class of time-series SCM (TiMINo), does not identify which mechanism changed, and has no fairness analysis.

Kostrzewa et al. [2026] repair counterfactuals as the classifier updates, instead of making a prediction about the future, keeping a set of counterfactuals valid and plausible across time. A counterfactual is repaired when it becomes invalid or its target-class confidence is low, and it is retired once the updated classifier already gives the original instance the target class. Repairing is cheaper than regenerating, although repaired counterfactuals can move further from the original instance. The main limitation of this work is that it is a proof of concept: it uses a local linear surrogate, does not model causal relationships, has no fairness mechanisms, and covers only gradual drift on low-dimensional synthetic datasets.

Upadhyay et al. [2021] propose ROAR, a framework that hardens recourse in advance against a bounded model shift through a minimax objective, and prove that the extra cost of this robustness is bounded by a 1-Lipschitz function. Unfortunately, the framework applies only to linear models (non-linear models are first approximated using LIME) and to non-causal recourse. It also assumes bounded shifts rather than continuous drift.

Rawal et al. [2020] give this problem a theoretical basis. They establish a lower bound on the probability that recourse is invalidated by a model update, and show a trade-off between that probability and recourse cost. Experiments on temporal, geospatial and data-correction shifts confirm that invalidation happens in practice, and that causally generated recourse is not immune to it. The work offers no solution, however, and its theory covers only sparse counterfactual recourse.

2.2.4.2 Recourse That Causes Its Own Drift

König et al. [2025] define performative validity as whether recourse still holds under the best model after people act on it. Their work proves that recourse can fail when actions intervene on, or are influenced by, effect variables. Improvement-focused causal recourse stays valid across a broad range of settings, whereas standard counterfactual and causal recourse often do not. The main limitations are that the work considers a single step, without repeated applications, that it assumes a single, static SCM, and that it requires causal knowledge.

Liu et al. [2025], on the other hand, follow the loop over many rounds with a limited number of positive decisions. Recourse pushes decision standards up, raising cost and failure rates. Fair-top-k and Dynamic Continual Learning reduce the damage, with recourse costs falling by around 70%: the former encourages diversity in the feature space by favouring people from less crowded regions, and the latter reduces how sharply the model changes. Here, there is no causal model, and the theory focuses on logistic regression in a saturated system. In both papers, drift comes from recourse and not from the outside world.

2.2.4.3 Learning Causal Structure from Change

Huang et al. [2020] propose Constraint-based causal Discovery from heterogeneous/NOnstationary Data (CD-NOD), a framework that leverages distribution

shifts to learn causal structure. It also extracts a low-dimensional "driving force" of the change and allows edges to appear and vanish over time. Perry et al. [2022] make sparsity explicit through the sparse mechanism shift hypothesis and propose the Mechanism Shift Score, proving that the full graph can be identified with high probability if the hypothesis holds. Although promising, both works are limited in scale: they are batch methods, so neither fits a data stream directly, and the kernel-based tests used by CD-NOD have a cost of $O(N^3)$ in the number of samples. Both also rule out most hidden confounders, sparse shift remains a hypothesis rather than an established fact, and neither work produces counterfactuals or recourse.

2.2.4.4 Tracking Causal Models Online

2.2.5 Detecting and Explaining Concept Drift

2.2.6 Fairness Under Distribution Shift

2.2.7 Interpretable Additive Models with Interactions

2.2.8 Readable Structural Equations

Ma et al. [2025] propose Causal-KAN, which puts SCM functions into KANs and encodes causal direction through initialisation and constraints, claiming less overfitting and more readable mappings. Alongside it, Almodóvar et al. [2026] show that KANs can be used to learn the structural equations of a known causal graph, so that the learned mechanisms can be inspected directly. Their method assumes causal sufficiency, and the authors note that it cannot guarantee that the learned mechanisms coincide with the true structural equations.

Additive structural equations also have a statistical precursor in Bühlmann et al. [2014], whose causal additive models (CAM) separate the search for a variable order from the sparse selection of edges, with consistency results in both low and high dimensions. CAM assumes that there are no hidden variables or cycles, and the authors note that such models are not closed under marginalisation, so a hidden variable can take the data outside the model class.

Nevertheless, none of these works provides any connection to recourse, much less to drift and fairness.

2.2.9 Comparison and Research Gaps

Work	Rec.	Causal	Fair	Time	Loc.	Read.	Unc.	Online
Counterfactual search								
Russell [2019]	✓							
Kanamori et al. [2020]	✓							
Karimi et al. [2020a]	✓							
Van Looveren and Klaise [2020]	✓							
Mothilal et al. [2020]	✓	(✓)						
White and d'Avila Garcez [2019]	✓							
Dandl et al. [2020]	✓							
Plausibility and causal consistency								
Poyiadzi et al. [2020]	✓							
Olson et al. [2021]	✓							
Joshi et al. [2019]	✓	(✓)						
Mahajan et al. [2019]	✓	(✓)						
Karimi et al. [2021]	✓	✓						
Karimi et al. [2020b]	✓	✓					✓	
Fairness of recourse								
Sharma et al. [2020]	✓		✓					
von Kügelgen et al. [2022]	✓	✓	✓					
Fonseca et al. [2023]	✓			✓				
Bell et al. [2025]	✓		✓	✓				
Recourse and causal models under drift								
De Toni et al. [2025]	✓	✓		✓			(✓)	
Kostrzewa et al. [2026]	✓			✓				(✓)
Upadhyay et al. [2021]	✓			(✓)				
Rawal et al. [2020]	✓	(✓)		✓				
König et al. [2025]	✓	✓		(✓)				
Liu et al. [2025]	✓		(✓)	✓				
Huang et al. [2020]		✓		✓	✓			
Perry et al. [2022]		✓		(✓)	✓			
Readable structural equations								
Ma et al. [2025]		✓				✓		
Almodóvar et al. [2026]		✓				✓		
Bühlmann et al. [2014]		✓				✓		
This work	○	✓	○	✓	✓	✓	✓	○

Table 2.2: Comparison of the reviewed works. **Rec.**: produces counterfactuals or recourse actions; **Causal**: uses a causal graph or structural causal model; **Fair**: measures or enforces fairness; **Time**: handles change over time (model, population or mechanism); **Loc.**: identifies which mechanism changed; **Read.**: learned mechanisms can be read directly; **Unc.**: quantifies uncertainty; **Online**: updates from a data stream. ✓ addressed; (✓) partly addressed; ○ planned in this work; blank: not addressed.

Chapter 3

Proposed Approach

3.1 Problem Formulation

3.2 Candidate Architectures

3.3 Experimental Design

3.4 Summary

Chapter 4

Conclusion

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Appendices

Appendix A

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Appendix B

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